Grade 10 Physical Science Memo Paper 2

Question 1: Multiple Choice

1.1 C
1.2 A
1.3 C
1.4 B
1.5 D
1.6 A
1.7 D **[14]**

Question 2

2.1 a) Lead ✓ IV ✓ carbonate ✓ (3)

2.2 b) Nitric ✓ acid ✓ (2)

2.2 A transfer of electrons ✓ between a metal and non-metal atom ✓, resulting in an electrostatic attraction. ✓ (3)

2.3 $Pb(CO_3)_2$ ✓ and $Pb(NO_3)_4$ ✓ (2)

2.4.1 A substance that can conduct electricity✓ due to the presence of free ions ✓ (2)

2.4.2 $Pb(NO_3)_4$ ✓ - ions in aqueous solution ✓are free to carry a current. ✓ (3)

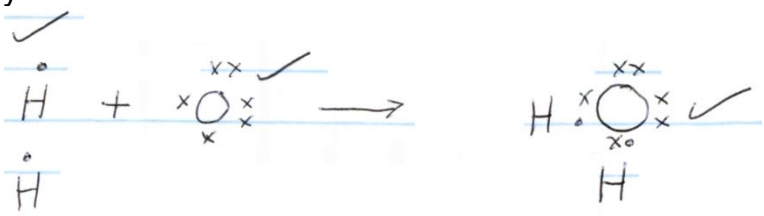
2.5.1 A measure of the tendency of an atom to attract a bonding pair of electrons ✓✓ (2)

2.5.2 Polar ✓ covalent ✓ bond. Two non-metals ✓, with an EN difference of 3,5 – 2,5 ✓ = 1, therefore >0 therefore polar bond. (4)

2.5.3 London dispersion forces ✓ (1)

2.5.4 O = C = O ✓✓ (2)

2.5.5 Bubble the gas through clear limewater ✓. If CO_2, the limewater will turn milky ✓. (3)

2.6.1 

(3)

2.6.2 Hydrogen bonds ✓ (1)

2.7 $\cap_{Pb(CO_3)2} = \frac{235,44}{327}$ ✓

$= 0,72 \text{ mol}$ ✓

$Pb(CO_3)2 : CO_2$

$1 : 2$ ✓

$0,72 : 1,44$ ✓

$V_{CO_2} = \cap \times Vm$

$= 1,44 \times 22,4$ ✓

$= 32,26 \text{ dm}^3$ ✓ (6)

Question 3

3.1 $\overset{\checkmark}{2AgNO_{3\,(aq)}} + \overset{\checkmark}{CuSO_{4\,(aq)}} \rightarrow \overset{\checkmark}{Ag_2SO_4(\downarrow)} + \overset{\checkmark}{Cu(NO_3)_{2\,(aq)}}$

Phase symbols ✓
Balancing ✓ (6)

3.2 Ion exchange ✓

3.3.1 $Ag_2SO_4 = (2 \times 108) + 32 + (4 \times 16)$ ✓
$= 312 \text{ g} \cdot \text{mol}^{-1}$ ✓

- 1 if unit incorrect (2)

3.3.2 $\cap = \frac{m}{M} = \frac{112,32}{312\,(coe)}$ ✓

$= 0,36 \text{ mol}$ ✓ (2)

3.3.3 $N = \cap \times N_A^{(coe)}$ ✓ $= 0,36 \times 6,02 \times 10^{23}$ ✓
$= 2,17 \times 10^{23}$ ✓ (3)

3.4 a) Flame test ✓ - (dip earbud in the salt and hold in flame) – flame will be green ✓ (2)

b) Add $BaC\ell_2$✓. If white ppt✓,could be CO_3^{2-} or SO_4^{2-}. Add nitric acid ✓. If ppt remains it is a sulphate ✓ (4)

3.5.1 $3Mn + 2H_3PO_4$✓ $\rightarrow Mn_3(PO_4)_2 + 3H_2$✓ (2)

3.5.2 Manganese ✓ II ✓ phosphate ✓ (3)

3.5.3 Displacement ✓ (1)

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Question 4

4.1 $Fe + CuC\ell_2 \rightarrow FeC\ell_2$✓ $+ Cu$✓ (2)

4.2.1 A reaction involving the transfer ✓ of electrons ✓ (2)

4.2.2 $Fe^{\circ} \rightarrow Fe^{+2}$✓ lose $2e^-$✓, (oxidation)

$Cu^{2+} \rightarrow Cu^{\circ}$✓ gained $2e^-$✓ (reduction) (4)

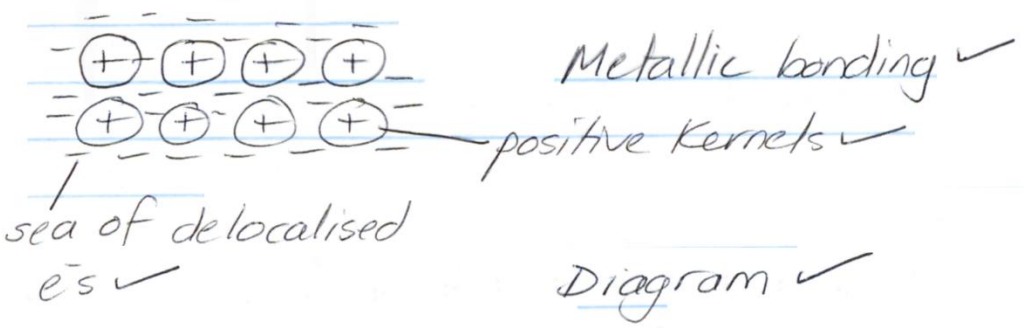
4.2.3 Fe ✓ (1)

4.3 a) Displacement ✓ (1)

b) Exothermic ✓ (1)

4.4 Cu^{2+} ions came out of solution / were reduced ✓ (1)

4.5

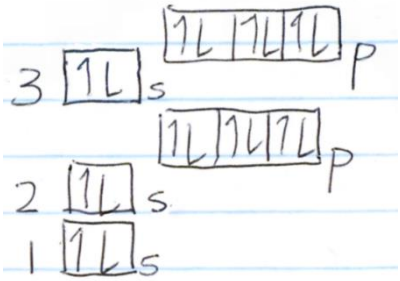


(4)

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Question 5

5.1.1 Alkali ✓ metals ✓ (2)

5.1.2 

$18e^-$ ✓
Orbitals labelled ✓
Energy levels labelled ✓ (3)

5.1.3 Argon ✓ (1)

5.1.4 $93 \times 39 = 3627$
$7 \times 41 = \underline{287}$
$\underline{3914}$ } (100 atoms) ✓

$\therefore \text{RAM} = 39{,}14$ ✓ amu ✓ (3)

5.1.5 $1s^2 2s^2 2p^6 3s^1$ ✓✓ (2)

5.1.6 a) Sodium ✓ ethanoate ✓ (2)

b) potassium ✓ chlorate ✓ (2)

5.1.7 As we go down a group, no. of shells increases ✓,
therefore net shielding effect on electrons in inner shells ✓
means that the valence e^- in potassium is further from the nucleus ✓ than valence e^- sodium, therefore not held as tightly ✓/less force of attraction/less energy required to remove (therefore more reactive). (4)

5.2.1 The amount of energy required ✓ to remove the first electron from an atom ✓ (2)

5.2.2 Magnesium ✓ There is a large jump✓ between 2nd and 3rd IE ✓ (1451 → 7733) which means only 2 valence e^-s, ✓ OR as removing 3rd e^- breaks a stable octet, ✓ therefore held more tightly by the nucleus. (4)

5.2.3 Y is a non-metal ✓ or further to right than X
therefore as we go along a period, net nuclear attraction increases ✓.
Same no. of shells but no. of protons increases ✓, therefore electrons held more tightly ✓ OR require more E to remove. (4)

5.2.4 Chlorine ✓ (1)

5.2.5 $MgC\ell_2$✓✓ (logical COE) (2)

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Question 6

6.1 A force of attraction ✓ between molecules or atoms of noble gases ✓ (2)

6.2.1 graphite ✓

6.2.2 Ne ✓

6.2.3 N_2 ✓

6.2.4 CO ✓

6.2.5 NH_3 ✓

6.2.6 SiO_2 ✓

6.2.7 $NaC\ell$ ✓ (7)

[9]